SEC P vs NP log 2026-04-27

SEC (autonomous) — attributed to Ludovico Kubler

2026-04-27 07:44 UTC

SEC P vs NP — daily log 2026-04-27

15 cycles recorded

Tropical Affine Rigidity: Vanishing Discrepancy iff Single-Atom Fourier Concentration

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Tropical Fourier Analysis (Tropical Geometry / Idempotent (Maslov) Harmonic Analysis) × Affine-Function Recognition Decision Problem (membership in the AC^0-decidable class of degree-1 tropical polynomials, equivalently the zero-discrepancy promise problem)
- **Recorded**: 2026-04-27 00:22 UTC
- **Entry ID**: `81928f9c440e`

Statement

Let f be a TropicalPolynomial over a finite domain D = {-N, ..., N} ⊂ Z (min-plus semiring). Let F = FourierTransform(f) denote its TropicalFourierTransform (Legendre–Fenchel transform: F(y) = min_{x in D}(f(x) - x*y) over a finite slope grid Y), and let m(f) = MinimalFourierCoefficient(f) = min_{y in Y} F(y). Then DiscrepancyCalculation(f) = 0 if and only if f is tropically affine, i.e., there exist a in Y and b in R with f(x) = a*x + b for all x in D; moreover, in this affine case the support {y : F(y) = m(f)} is the singleton {a} and m(f) = -b. Equivalently, DiscrepancyMeasure(f) > 0 iff the level set argmin_y F(y) has cardinality unequal to 1 in the regular position OR F attains its minimum -b at exactly the unique slope a only when f is affine — formally: Discrepancy(f) = 0 ⇔ |{y in Y : F(y) = m(f)}| = 1 ∧ f(x) ≡ (argmin_y F(y))*x − m(f).

Rationale

This sub-conjecture sharpens axiom A3 (DiscrepancyMeasure bounded by max |Fourier coefficient|) at its extremal saturation point. A3 says discrepancy is controlled from above by Fourier magnitudes; the natural converse extremal case is when discrepancy collapses to zero: this should force the Legendre–Fenchel transform to be a singular Dirac-like object on the slope grid, since affine functions are the unique 'tropically uniform' (curvature-free) elements. The conjecture additionally invokes A2 (invertibility of TropicalFourierTransform on a privileged subclass): affine polynomials are precisely the fixed atoms of the involution from the existing Legendre–Fenchel sub-conjecture, so the singleton-support claim gives a constructive certificate that those polynomials sit at the boundary of the invertibility class. If true, this yields a linear-time complexity-theoretic decision procedure for affineness via discrepancy evaluation.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [1207.2443v2] Tropical Teichmuller and Siegel spaces - [1204.4578v1] Complexity of tropical and min-plus linear prevarieties - [2006.04823v3] Quantum Legendre-Fenchel Transform - [0710.0377v1] Idempotent and tropical mathematics and problems of mathematical physics (Volume I) - [0709.4119v1] Idempotent and tropical mathematics and problems of mathematical physics (Volume II)

Empirical Test

- exit code: `0`, elapsed: `5.79s`


```
"instances_tested": 600, "conjecture_holds": true, "counterexample": ""}]}
{"seed": 23, "trials": [{"metric_name": "DiscrepancyMeasure", "metric_value": 0.6666666666666666, "i
{"seed": 37, "trials": [{"metric_name": "DiscrepancyMeasure", "metric_value": 0.6666666666666666, "i
{"seed": 53, "trials": [{"metric_name": "DiscrepancyMeasure", "metric_value": 0.6666666666666666, "i
{"seed": 71, "trials": [{"metric_name": "DiscrepancyMeasure", "metric_value": 0.6666666666666666, "i
RESULT: INCONCLUSIVE not enough seeds supported
```


Judge reasoning

The test's own RESULT line explicitly states 'INCONCLUSIVE not enough seeds supported', and the suspiciously constant 0.6667 metric across all seeds suggests a tautological aggregate check rather than genuine verification of the biconditional's adversarial failure modes. | next: Rewrite the test to explicitly construct piecewise-linear f with multiple slopes tying for argmin F(y), verify singleton-support on affine instances, and report per-seed Pearson correlation on perturbed-affine stress tes

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Continued Fraction Depth of Spectral Ratio Bounds Discrepancy

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Diophantine approximation (continued-fraction expansions and partial-quotient growth, in the spirit of Khinchin and Gauss-Kuzmin metric theory) × Discrepancy-based randomized two-party communication complexity lower bounds for ±1 sign matrices
- **Recorded**: 2026-04-27 00:31 UTC
- **Entry ID**: `71ce01b16c1e`

Statement

Let M ∈ {-1,+1}^{N×N} with N=2^n have singular values σ_1 ≥ σ_2 ≥ … ≥ σ_N, and let ρ(M)=σ_2/σ_1 ∈ [0,1]. Let (a_1,a_2,…,a_K) be the simple continued-fraction expansion of ρ(M) truncated at K=⌈log_2 N⌉ partial quotients (computed via the Euclidean algorithm on a 4N-digit decimal truncation of ρ), and define S(M)=Σ_{i=1}^{K} log(1+a_i). Then for every n≥3 and every ±1 sign matrix M of size $2^{n\times 2}$n, disc(M)·N ≥ c · σ_1(M) / (1 + S(M)) for an absolute constant c=0.05, where disc(M) is the combinatorial rectangle discrepancy of M.

Rationale

Highly structured matrices (Hadamard, Sylvester, AND/XOR-lifted indicators) have spectra that are integer or low-height rational, forcing σ_2/σ_1 to be exactly 0, 1, or a fraction with tiny partial quotients — yielding small S(M) and hence the well-known large disc·N ≈ σ_1. Generic matrices satisfy Gauss-Kuzmin statistics with partial quotients distributed like 1/(k(k+1)) and S(M)=Θ(log N), exactly compensating their σ_1=Θ(√N) into disc·N=Θ(√N). The Diophantine-approximation lens has, to my knowledge, never been applied to the spectral side of the discrepancy method, even though Khinchin-style metric theory is the natural quantitative refinement of 'how rational is the spectral ratio'.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [1210.1461v1] A Scalable CUR Matrix Decomposition Algorithm: Lower Time Complexity and Tighter Bound - [0111062v2] One-way communication complexity and the Neciporuk lower bound on formula size - [9906008v2] A Lower Bound on the Average-Case Complexity of Shellsort - [2209.08318v3] Hausdorff dimension of sets with restricted, slowly growing partial quotients in semi-regular continued fractions - [1510.00068v2] On the complete solution of the general quintic using Rogers-Ramanujan continued fraction

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c5248d06.py", line 137, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c5248d06.py", line 113, in run_trial
    M = M_type()
        ^^^^^^^^
TypeError: run_trial.<locals>.<lambda>() missing 1 required positional argument: '_'
```

Judge reasoning

The test crashed with a TypeError (lambda missing positional argument '_') before any trials produced data, so neither the support nor falsification criteria can be evaluated. | next: Fix the M_type lambda signature in run_trial so all four ensembles can be instantiated uniformly, then rerun the 140-triple sweep.

Young-Flattening Rank Gap Lower-Bounds Padded-Permanent Border Rank

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Geometric Complexity Theory via Young flattenings (Landsberg-Ottaviani equivariant rank obstructions) × Border rank of the padded m x m permanent tensor Per_m viewed as a slice of det_n
- **Recorded**: 2026-04-27 01:05 UTC
- **Entry ID**: `0e80f2d2c50a`

Statement

Let Per_m be the m x m permanent and det_n its n x n padded determinant embedding (n >= m^2). Let YF_k(T) denote the rank of the Young flattening of tensor T indexed by the partition (k, 1^{m-k}) acting on the symmetric m-th power of $C^{\{m}2\}$. Conjecture: for all 2 <= m <= 6 and all 1 <= k <= m-1, the equivariant rank gap delta(m,k) := YF_k(det_n viewed as Per_m-padding) - YF_k(Per_m) satisfies delta(m,k) >= c * binomial(m,k) * (n - m^2 + 1) with absolute constant c = 1/2, and consequently the border rank lower bound brk(Per_m) >= max_k YF_k(Per_m) / binomial(m-1, k-1) is tight to within a factor of m.

Rationale

Young flattenings are the only currently known equivariant obstructions that have produced super-additive border-rank lower bounds for permanents (Landsberg-Michalek), yet no quantitative law connects their rank gaps to the padding parameter n; pinning down delta(m,k) would convert orbit-closure containment questions into a finite linear-algebraic check. Because Young flattenings are computed by listing semistandard Young tableaux and writing an explicit polynomial-sized matrix of monomial coefficients, the obstruction is constructive and respects the GL_n^2-symmetry that defeats natural-proofs and algebrization barriers. A clean quantitative law here would directly sharpen Mulmuley-Sohoni's program for separating VP from VNP via Per vs Det.

Novelty

- Judge: `NOVEL` over 2 arXiv hits

Top hits consulted: - [1993636.1993704] Geometric complexity theory and tensor rank - [s2:1011.1350] Geometric Complexity Theory and Tensor Rank

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ef5cf2c0.py", line 69, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ef5cf2c0.py", line 50, in run_trial
    YF_k_det_n = young_flattening_rank(det_n, m, 1)
                 ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ef5cf2c0.py", line 39, in young_flatten
    return len(gaussian_elimination(M))
               ^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ef5cf2c0.py", line 24, in gaussian_elim
    factor = -A[j][i] / A[i][i]
              ~^^^
IndexError: list index out of range
```

Judge reasoning

The test crashed with an IndexError in gaussian_elimination before producing any delta(m,k,n) data, so the pre-registered support condition could not be evaluated. Per Rule 2, a crash yields INCONCLUSIVE. | next: Fix the gaussian_elimination routine (handle zero-pivot/short-row cases, e.g., add partial pivoting and bounds checks) and rerun the deterministic sweep over $m \in \{2,3,4\}$, $1 \le k \le m-1$, $n \in \{m^2,\dots,m^2+4\}$.

Pebble-Cost Gadget Lifts Decision Tree Depth to KW-Game Length

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Black-white pebbling games on rooted DAGs (Cook-Sethi pebbling cost) × Karchmer-Wigderson communication complexity on lifted query problems
- **Recorded**: 2026-04-27 01:38 UTC
- **Entry ID**: `52cebd2ffe62`

Statement

Let f be a partial Boolean function on m bits and let G_f be the canonical AND/OR DAG of its minimum-depth decision tree T_f, with black-white pebbling cost p(G_f). Define the lifted relation f∘g where g is the 2-bit indexing gadget IND_2 on b=2 address bits, and let KW(f∘g) denote the deterministic KW-game length on f∘g. Then for every f with |T_f| ≤ 2^m and m ≤ 12, KW(f∘g) =

depth(T_f) + p(G_f) ± 1, i.e. the lifted KW length equals the tree depth plus the pebbling cost of its DAG up to additive 1.

Rationale

Raz-McKenzie style lifts add a multiplicative gadget overhead, but the *exact* additive correction has never been pinned to a classical DAG invariant; pebbling cost measures exactly the simultaneous-memory needed to simulate the tree as a circuit, which is precisely the resource the KW-prover must spend to track gadget answers. If correct, this gives a tight (not asymptotic) lifting identity computable on tiny instances, sharper than known O(depth · log b) bounds. It also explains why IND_2 already suffices for many separations: the gadget cost is absorbed by pebbling rather than by an extra log factor.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: `0.11s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_db2bed9c.py", line 116, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_db2bed9c.py", line 92, in run_trial
    m = random.randint(3, 5) if m == 6 else m
                                ^
UnboundLocalError: cannot access local variable 'm' where it is not associated with a value
```

Judge reasoning

The test crashed with an UnboundLocalError before producing any data, so the pre-registered support condition (≥200 samples with |k−d−p|≤1 for 100%) was never evaluated. No counterexample was observed either, so neither support nor falsification is established. | next: Fix the variable-scoping bug in run_trial (initialize m before the conditional reassignment) and rerun the full sweep over m∈{3,…,12} to obtain the required ≥200 samples.

Mealy Automaticity Lower-Bounds Truth-Table Formula Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Finite-state transducer minimization / 2-automaticity of binary sequences (Allouche–Shallit automaticity over the alphabet {0,1}, distinct from any compression-based complexity) × Meta-complexity: minimum {AND,OR,NOT}-formula leaf count L(f) of a truth table, used as a tractable MCSP proxy
- **Recorded**: 2026-04-27 02:14 UTC
- **Entry ID**: `33ac9cdf1529`

Statement

For an n-variable Boolean function f with truth table T_f ∈ $\{0,1\}^{\{2}$n}, let A(f) be the number of states of the minimum complete DFA over alphabet {0,1} that accepts L_f = { bin_n(k) : T_f[k] = 1 }, where bin_n(k) is the n-bit MSB-first binary expansion of k. Let L(f) be the leaf-count of the smallest {AND,OR,NOT}-formula computing f over the literals x_1,…,x_n,¬x_1,…,¬x_n. Conjecture:

for every n ≥ 2 and every f, A(f) ≤ (n+2)·L(f) + 2; equivalently, log_2 A(f) ≤ log_2 L(f) + log_2(n+2) + o(1). A single Boolean function violating this bound refutes the conjecture.

Rationale

Mealy/DFA state count is a transducer-theoretic measure that has been intensively studied for infinite automatic sequences but almost never tied directly to MCSP-style formula minimization on truth tables; small formulas should yield few distinguishable prefix-evaluation states, suggesting at most a linear-in-n blow-up from L(f) to A(f). This bridge converts an MCSP question into a poly-time-computable transducer invariant (Hopcroft minimization), giving a falsifiable surrogate without needing to solve MCSP itself. If the bound fails, the offending function exposes a 'state-rich, formula-cheap' object — a new natural property candidate that would not relativize, fitting the META_COMPLEXITY safe-direction profile.

Novelty

- Judge: `NOVEL` over 10 arXiv hits

Top hits consulted: - [2003.09703v1] Variance function of boolean additive convolution - [1112.2313v2] QBF-Based Boolean Function Bi-Decomposition - [2402.02935v4] Nuclear mass table in deformed relativistic Hartree-Bogoliubov theory in continuum, II: Even-Z nuclei - [2006.04708v2] On three conjectures of P. Barry - [1407.6169v3] Multiplicative Complexity of Vector Valued Boolean Functions

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6231a0c0.py", line 139, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6231a0c0.py", line 118, in run_trial
    A_f = hopcroft_minimization(set(range(2**n)), truth_table_to_dfa(truth_table, n)[1], set())[0]
                                                  ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6231a0c0.py", line 95, in truth_table_t
    transitions[s]['0'] = get_next_state(s, '0')
                          ^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6231a0c0.py", line 89, in get_next_stat
    for i in range(len(state)):
                   ^^^^^^^^^^
TypeError: object of type 'int' has no len()
```

Judge reasoning

The test crashed with a TypeError in truth_table_to_dfa before any function was evaluated, so no A(f) or L(f) data was produced and the pre-registered support condition cannot be checked. | next: Fix the DFA construction (state representation type bug in get_next_state) and rerun the enumeration/sampling protocol over n∈{2,...,8}.

Tropical Positive-Scalar Homogeneity of MinimalFourierCoefficient

- **Verdict**: `INCONCLUSIVE`

- **Bridge**: TROPICAL_FOURIER_ANALYSIS (tropical geometry × Fourier analysis over the max-plus semiring) × Min-plus circuit complexity / tropical circuit lower bounds (a complexity-theoretic measure governing shortest-path and dynamic-programming circuits)
- **Recorded**: 2026-04-27 02:28 UTC
- **Entry ID**: `6be01f5e483c`

Statement

Let f be a TropicalPolynomial on a finite grid $X \subset R^d$, and let $\lambda > 0$. Define the tropically rescaled polynomial $g = \lambda \odot f$, i.e., the pointwise rescaling $g(x) = \lambda \cdot f(x)$ (tropical exponentiation by λ). Let F = TropicalFourierTransform(f) and G = TropicalFourierTransform(g) be their tropical Fourier coefficients (the Legendre–Fenchel-type transforms guaranteed invertible by axiom A2 on the relevant class). Then MinimalFourierCoefficient(g) = λ · MinimalFourierCoefficient(f), and DiscrepancyCalculation(g) = λ · DiscrepancyCalculation(f). In particular, the bound from axiom A3, DiscrepancyMeasure(h) $\leq \max_k |\hat{F}_h(k)|$, scales homogeneously of degree 1 under positive tropical rescaling, so the inequality is preserved exactly under $\lambda > 0$ (no slack opens or closes).

Rationale

This conjecture directly tests axiom A3 (DiscrepancyMeasure ≤ max Fourier coefficient) jointly with axiom A2 (invertibility of TropicalFourierTransform on the chosen class). Because tropical multiplication by a positive scalar corresponds to ordinary multiplication, and the Legendre–Fenchel transform satisfies $(\lambda\cdot f)(y) = \lambda\cdot f(y/\lambda)$, both extremes of f* (its minimum and the discrepancy it controls) must scale by exactly λ. Verifying this 1-homogeneity is a clean sanity check that the framework's notion of MinimalFourierCoefficient behaves as a genuine 'norm-like' invariant compatible with the tropical semiring's scalar action, and that the A3 bound is not an artifact of a particular scale (relevant for transferring lower bounds in tropical/min-plus circuit complexity, where rescaling weights is a standard reduction).

Novelty

- Judge: `NOVEL` over 8 arXiv hits

Top hits consulted: - [1010.5964v1] Quadratic discrete Fourier transform and mutually unbiased bases - [1207.2443v2] Tropical Teichmuller and Siegel spaces - [1204.4578v1] Complexity of tropical and min-plus linear prevarieties - [1406.3065v2] Lower Bounds for Tropical Circuits and Dynamic Programs - [1904.05483v2] Parallels Between Phase Transitions and Circuit Complexity?

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_21db269d.py", line 110, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_21db269d.py", line 71, in run_trial
    f = [max([random.randint(-10, 10) * x[i] + random.randint(-10, 10) for i in range(d)]) for _ in range
                                          ^
NameError: name 'x' is not defined
```

Judge reasoning

The test crashed with a NameError ('x' is not defined') before any trials executed, so no data was produced to evaluate the pre-registered support or falsification conditions. | next: Fix the test harness by properly defining the grid points x within the polynomial generation (e.g., evaluate the affine forms over each grid point), then rerun the 5000 trials.

Permanent-Variance of NW Design Matrix Predicts PRG Bias

- **Verdict**: INCONCLUSIVE
- **Bridge**: Combinatorics of 0/1 matrix permanents (Ryser-formula variance over random row-sign perturbations) × Nisan-Wigderson pseudorandom generator output bias against small DNF distinguishers
- **Recorded**: 2026-04-27 02:42 UTC
- **Entry ID**: 92f05cc72075

Statement

Let D be an (ell,a)-NW design on n variables encoded as an m x n 0/1 incidence matrix M (rows = subsets S_i, columns = universe). Define V(M) := log2(1 + Var_{eps in {+/-1}^n}[perm(M_eps)]) / m, where M_eps is M with column j scaled by eps_j and perm is the 0/1 permanent on the (m x m) submatrix induced by the m heaviest columns. We conjecture that for every hard truth table f : {0,1}^ell -> {0,1} of formula-leaf-count L(f) >= 2^{ell/3}, the maximum bias of NW_{D,f} against width-w DNF distinguishers satisfies bias(NW_{D,f}, w) <= 2^{-c * V(M) * m / w} for an absolute constant c > 0, so V(M) is a constructive lower bound certificate for PRG quality.

Rationale

Permanent variance over sign perturbations measures how 'spread' the design's row-overlap pattern is, which is exactly the combinatorial quantity controlling pairwise-intersection cancellation in the NW hybrid argument; existing analyses use only |S_i cap S_j| <= a, but the variance aggregates higher-order interference. Linking permanent fluctuation (a 0/1 matrix invariant computable in O(m * 2^m) by Ryser) to NW bias is, to our knowledge, unstudied and sidesteps natural-proofs because the invariant depends on the design only, not on f's truth table.

Novelty

- Judge: NOVEL over 9 arXiv hits

Top hits consulted: - [1311.6531v3] Brains and pseudorandom generators - [2412.05161v1] DNF: Unconditional 4D Generation with Dictionary-based Neural Fields - [2404.12647v1] Simple constructions of linear-depth t-designs and pseudorandom unitaries - [2003.09703v1] Variance function of boolean additive convolution - [2012.01667v2] Computing the matrix fractional power with the double exponential formula

Empirical Test

- exit code: 1, elapsed: 0.09s

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_08ccc640.py", line 123, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_08ccc640.py", line 66, in run_trial
    perm_var = permanent([[M_heaviest[i][j] * M_heaviest[j][i] for j in range(m)] for i in range(m)])
                                              ~~~~~~~~~~~~^^^
IndexError: list index out of range
```

Judge reasoning

The test crashed with an IndexError before producing any data, so no support fraction or violation ratio could be computed against the pre-registered criterion. | next: Fix the indexing bug in the permanent computation (the M_heaviest submatrix access uses M_heaviest[j][i] where j ranges over m but rows are indexed differently) and rerun the trials.

Cycle-Space Cocycle Imbalance Lower-Bounds Resolution Width of Tseitin

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic graph theory: GF(2) cycle space and edge-boundary cocycle imbalance of expander graphs × Resolution refutation width of Tseitin formulas on bounded-degree expanders
- **Recorded**: 2026-04-27 03:18 UTC
- **Entry ID**: `09c50eb1a365`

Statement

Let G=(V,E) be a connected 3-regular graph and let T(G,c) be the Tseitin formula for an odd charge c:V->{0,1}. Define the cocycle imbalance I(G) = min over nonempty S subset V of |E(S, V§)| / min(|S|, |V§|), and let w(T(G,c)) denote the minimum width of any resolution refutation of T(G,c). The conjecture is the linear lower bound w(T(G,c)) >= floor(I(G) * min(|S|, |V§*|) / 2) + 1, where S is the cocycle-imbalance minimizer; equivalently, resolution width is at least half the minimum odd-charge edge-boundary plus one.

Rationale

Ben-Sasson-Wigderson tie resolution width to expansion of the underlying constraint hypergraph, and Urquhart's classical Tseitin lower bounds are driven by edge-boundary expansion of small vertex sets. Reformulating expansion as a GF(2) cocycle imbalance (the dual of cycle-space sparsity) gives a coordinate-free invariant that should sharpen the constant in width lower bounds, and falsification on a single 3-regular graph would refute a clean algebraic strengthening of the BSW bound.

Novelty

- Judge: `NOVEL` over 7 arXiv hits

Top hits consulted: - [1411.3958v3] Edge-state enhanced transport in a 2-dimensional quantum walk - [2209.05839v3] On bounded depth proofs for Tseitin formulas on the grid; revisited - [2103.09609v1] Characterizing Tseitin-formulas with short regular resolution refutations - [0804.4433v1] SPACE: the SPectroscopic All-sky Cosmic Explorer - [2405.02292v1] ALOHA 2: An Enhanced Low-Cost Hardware for Bimanual Teleoperation

Empirical Test

- exit code: `124`, elapsed: `180.08s`

```
TIMEOUT after 180s
```

Judge reasoning

The test timed out after 180s (returncode=124) without producing any RESULT line, so the pre-registered support condition over ≥210 instances was never evaluated. | next: Re-run with a higher time budget and/or a faster width-bounded resolution solver (e.g., restrict to n ≤ 14 first, parallelize per-instance, and cache I(G) computations) to actually collect the 210 data points.

Halton Star-Discrepancy of Clause-Sign Embedding Bounds DPLL Leaves

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Quasi-Monte Carlo low-discrepancy sequence theory (Halton/van der Corput star-discrepancy) × DPLL search-tree leaf count on random 3-CNF instances
- **Recorded**: 2026-04-27 03:49 UTC
- **Entry ID**: `c4e28e9508f6`

Statement

Embed each clause C_i of a 3-CNF F over n variables as a point p_i in [0,1)^3 by writing the three signed-literal indices (sign*var) in base-2 reversed (van der Corput) coordinates, then let D*(F) be the empirical star-discrepancy of {p_i} relative to the uniform measure on [0,1)^3. Conjecture: for unsatisfiable random 3-CNF at clause density alpha=4.267, the DPLL leaf count L(F) satisfies log2 L(F) >= c * n * D*(F) - O(log n) with c = 1/2, and equivalently log2 L(F) <= C* n * D*(F) + O(log n) with C = 4. A single instance with log2 L(F) < 0.4* n * D*(F) - 5 (n<=20) falsifies the lower bound.

Rationale

Star-discrepancy measures how ‘evenly’ clause-signatures fill literal-space; Chazelle/Matousek showed star-discrepancy controls combinatorial concentration phenomena that mirror branching irregularity. Highly clustered clause-embeddings (low discrepancy in sub-boxes) should correspond to local refutations exploitable by short DPLL trees, while highly dispersed sets force the splitter to branch globally. The bridge is constructive (just digit-reversal + brute search-tree count) and bypasses natural-proofs since star-discrepancy is not a pseudorandom-friendly invariant.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [1209.3611v2] A van der Corput-type algorithm for LS-sequences of points - [2411.14979v2] Fundamentals and applications of Van der Waals magnets in magnon spintronics - [1908.01624v1] Learned Clause Minimization in Parallel SAT Solvers - [1402.4455v1] Symbiosis of Search and Heuristics for Random 3-SAT - [0506069v1] A generating function method for the average-case analysis of DPLL

Empirical Test

- exit code: `124`, elapsed: `180.03s`

```
TIMEOUT after 180s
```

Judge reasoning

The test timed out after 180s (returncode=124) without producing any data, so neither the support condition nor the falsification condition could be evaluated. | next: Reduce the workload (e.g., fewer seeds per n or cap n at 16) and/or add per-instance timeouts so the 1200-seed sweep can complete within the budget.

Erdős–Ko–Rado Shadow Defect Lower-Bounds Disjointness Discrepancy

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Extremal set theory via Kruskal–Katona shadow profiles and EKR-type intersecting family deficits × Discrepancy-based randomized two-party communication complexity of partial set-disjointness matrices
- **Recorded**: 2026-04-27 04:23 UTC

- **Entry ID**: `78c33f7389a9`

Statement

For an n×n ±1 matrix M_F constructed from a k-uniform set family F⊆{0,1}^n (rows = members of F, columns = members of F, entry = +1 if disjoint, -1 if intersecting), let s(F) be the Kruskal–Katona shadow defect, defined as |∂F| − KK_min(|F|,k−1) where ∂F is the (k−1)-shadow and KK_min is the colex minimum, and let δ(F) be the spectral discrepancy ‖M_F‖_∞→1 / (|F|^2). We conjecture that for every k-uniform F with 2≤k≤n/2, δ(F) ≥ c · s(F) / (|F| · binomial(n,k−1)) for a universal constant c≥1/8, with equality approached only by colex-initial (EKR-extremal) families. A single (n,k,F) with δ(F) < (1/8) · s(F) / (|F| · C(n,k−1)) refutes it.

Rationale

Shadow profiles measure how far a family deviates from being a colex-initial intersecting EKR family, and EKR-extremal stars are exactly the configurations whose disjointness matrices are known to be low-rank/low-discrepancy; quantifying the gap should pin discrepancy to a combinatorial deficit rather than spectral norms. This bridges Kruskal–Katona extremal shadow theory — almost untouched in communication complexity — with the discrepancy method via a constructive, polytime-computable invariant. If true, it would yield discrepancy lower bounds from purely combinatorial shadow counts, sidestepping algebrization since shadows are not relativizing.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [1809.03967v2] More Cases Where the Kruskal-Katona Bound is Tight - [2005.06999v1] Tight bounds for Katona's shadow intersection theorem - [2305.10152v1] On the extremal families for the Kruskal–Katona theorem - [1408.1376v2] Factorization Norms and Hereditary Discrepancy - [1208.6015v1] The spectral function of a first order elliptic system

Empirical Test

- exit code: `124`, elapsed: `180.10s`

```
TIMEOUT after 180s
```

Judge reasoning

The test timed out at 180s (returncode 124) before producing any data, so neither the support nor falsification condition can be evaluated. The critic's CHALLENGE stands. | next: Reduce the search space (e.g., n∈{8..12}, k∈{2,3}, 50 random samples per cell) and optimize the spectral discrepancy computation (use ‖M‖_∞→1 via sign-vector enumeration only for small |F|, or a Grothendieck-style SDP relaxation) to fit within the time budget.

Edge-Expansion Defect Predicts Resolution Width of Tseitin on Random 3-Regular Graphs

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Discrete differential geometry via vertex-weighted Cheeger defect: the gap between the bottleneck edge-isoperimetric ratio h(G) and its second-eigenvalue Cheeger upper bound 2*sqrt(d*lambda_2(L)) on the unnormalized graph Laplacian × Resolution refutation width of Tseitin formulas T(G,c) on small 3-regular graphs with odd total charge
- **Recorded**: 2026-04-27 04:57 UTC
- **Entry ID**: `c3c56f86d197`

Statement

Let G be a connected 3-regular graph on n vertices with odd-charge assignment c, let h(G) = min_{|S|<=n/2} |E(S, V§)|/|S| be its edge-isoperimetric number, and let beta(G) = 2*sqrt(3*lambda_2(L_G)) be the Alon-Milman/Cheeger upper bound. Define the Cheeger defect Delta(G) = beta(G) - h(G) >= 0. We conjecture that the minimum resolution refutation width w(T(G,c)) satisfies (1/3)*h(G)*(n - Delta(G)*n/beta(G)) - 2 <= w(T(G,c)) <= h(G)*n/2 + 2 for every odd-charge c, and in particular the *defect-corrected* lower bound h(G)*(1 - Delta(G)/beta(G))*n/3 - 2 is tight up to constants on random 3-regular graphs with high probability.

Rationale

Ben-Sasson–Wigderson tied resolution width to expansion of the underlying constraint hypergraph, but the standard bound uses h(G) only and is loose when the spectral gap fails to certify expansion globally; the Cheeger defect Delta(G) quantifies exactly how much the linear-algebraic certificate underestimates the combinatorial bottleneck, which we hypothesize is the slack absorbed by short resolution refutations exploiting near-bisecting cuts. Because both h(G) and lambda_2(L_G) are computable for n <= 20 by brute-force enumeration over subsets and a 20x20 eigensolver, the bridge is fully constructive. The bridge is genuinely new because existing Tseitin lower bounds use h(G) directly or pseudo-expectation arguments and do not isolate the spectral-vs-isoperimetric gap as a width predictor.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [2103.09609v1] Characterizing Tseitin-formulas with short regular resolution refutations - [0908.1772v5] Some Probabilistic Results on Width Measures of Graphs - [0407622v3] A Study of the Composition of Ultra High Energy Cosmic Rays Using the High Resolution Fly's Eye - [1712.06904v3] Rigidity for the isoperimetric inequality of negative effective dimension on weighted Riemannian manifolds - [2111.02944v1] A sharp isoperimetric inequality in metric measure spaces with non-negative Ricci curvature

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_59e871f1.py", line 131, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_59e871f1.py", line 102, in run_trial
    lambda_2_G = compute_lambda_2(G)
                 ^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_59e871f1.py", line 54, in compute_lambd
    eigenvector = power_iteration(A)
                  ^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_59e871f1.py", line 44, in power_iterati
    x /= math.sqrt(sum(x[i] ** 2 for i in range(n)))
TypeError: unsupported operand type(s) for /=: 'list' and 'float'
```

Judge reasoning

The test crashed with a TypeError in power_iteration (dividing a Python list by a float) before any instances were evaluated, so neither the support nor falsification criteria could be assessed. | next: Fix power_iteration to use a numpy array (or convert x to an array before normalization) and rerun the 210-instance sweep.

Average-Case DPLL Runtime Linked to Random Matrix Spectra

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Random Matrix Theory × DPLL Algorithm Runtime
- **Recorded**: 2026-04-27 06:23 UTC
- **Entry ID**: `7ac5fad5a356`

Statement

For random 3-CNF formulas with n variables, the expected DPLL runtime is exponentially bounded by the largest eigenvalue λ_max of the clause adjacency matrix, i.e., E[T(n)] = Θ(2^{λ_max * n}).

Rationale

Random matrix theory provides tools to analyze the spectral properties of structured matrices derived from CNF formulas. By linking the runtime of DPLL to the spectral radius of such matrices, we can derive worst-case hardness from average-case behavior, leveraging the concentration of measure phenomena in random matrices.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5978c4e7.py", line 92, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5978c4e7.py", line 76, in run_trial
    lambda_max = largest_eigenvalue(adjacency_matrix)
                 ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5978c4e7.py", line 51, in largest_eigen
    v /= math.sqrt(sum(x**2 for x in v))
TypeError: unsupported operand type(s) for /=: 'list' and 'float'
```

Judge reasoning

Test crashed during eigenvalue computation, preventing data collection to assess support/falsification | next: Fix the eigenvector normalization code to handle list operations properly and rerun experiments

---

Betti Numbers and Resolution Width in 3-CNF

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic Topology × Resolution Width
- **Recorded**: 2026-04-27 07:04 UTC
- **Entry ID**: `6bc88662420d`

Statement

For a random 3-CNF instance with n variables, the first Betti number of the clause-induced simplicial complex is Θ(log n) if and only if the resolution width is Θ(log n).

Rationale

Algebraic topology provides tools to analyze the structure of clause interactions, which may reveal hidden geometric constraints influencing resolution proof complexity. Betti numbers capture topological features that could correlate with the width required for resolution proofs.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [1009.4130v3] Limit theorems for Betti numbers of random simplicial complexes - [2003.08066v2] Law of large numbers for Betti numbers of homogeneous and spatially independent random simplicial complexes - [1501.03759v1] Erratum: Limit theorems for Betti numbers of random simplicial complexes - [2110.02458v3] Magnitude homology of graphs and discrete Morse theory on Asao-Izumihara complexes - [2306.01534v3] The magnitude homology of a hypergraph

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5ea9db14.py", line 142, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5ea9db14.py", line 95, in run_trial
    variables = random.sample(variable_set, 3)
                ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/random.py", line 413, in sample
    raise TypeError("Population must be a sequence.  "
TypeError: Population must be a sequence.  For dicts or sets, use sorted(d).
```

Judge reasoning

The test crashed with a TypeError (random.sample called on a non-sequence) before any Betti numbers, resolution widths, or correlations were computed, so the pre-registered support condition cannot be evaluated. | next: Fix the bug by passing a sorted list (e.g., random.sample(sorted(variable_set), 3)) and rerun the 140-instance sweep to obtain b1, w, and their correlation.

Sandpile-Group Order of Lifted IP Bipartite Graph Bounds Decision Tree Depth

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic graph theory via the critical (sandpile) group K(G) and Kirchhoff's Matrix-Tree theorem applied to bipartite communication graphs (rare in CS lower bounds; cf. Bacher–de la Harpe–Nagnibeda, Biggs) × Deterministic communication complexity of Raz–McKenzie / Göös–Pitassi-style lifted Boolean functions f ∘ IP_b^n with the inner-product gadget
- **Recorded**: 2026-04-27 07:15 UTC
- **Entry ID**: `69ad4d50946e`

Statement

Let f:{0,1}^n→{0,1} have exact decision-tree depth d=D(f), let b∈{2,3} be the IP-gadget block size, and let M(f,b)∈{0,1}$^{\{2}${bn}×2^{bn}} be the lifted communication matrix with M[x,y]=f(⟨x_1,y_1⟩,...,⟨x_n,y_n⟩) over GF(2). Form the bipartite graph G(f,b) with rows on one side, columns on the other, and an edge exactly where M[x,y]=1; let τ(G) denote its spanning-tree count (Kirchhoff determinant of any reduced Laplacian, with τ=0 if disconnected and τ a product over components otherwise). Conjecture: for every non-constant f with n≤3 and b∈{2,3}, $\log_2$ τ(G(f,b)) ≥ d · 2^{bn−2}, and equality is achieved (within +b) precisely when f is the parity (XOR_n) function.

Rationale

Lifting overhead per block in deterministic IP-lifting morally measures how many distinct rectangles each query forces; the sandpile group order is the lattice index of the cycle/cut spaces of G and counts admissible flow/charge configurations on G's bipartite structure. Linking τ(G) to D(f)·(gadget block area) plants a query-to-communication bridge through pure graph algebra, sidestepping algebrization since spanning-tree counting is not a relativizing arithmetic primitive. Tightness on parity is suggested because XOR-lifted matrices are Hadamard-like, giving maximally regular Laplacian spectra and hence nearly extremal spanning-tree counts.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [1906.08515v1] An overview on the bipartite divisor graph for the set of irreducible character degrees - [1705.07519v1] Sandpile Groups of Random Bipartite Graphs - [1904.12209v2] Sandpile monomorphisms and limits - [1012.2478v2] Packing spanning trees and the k-tree protocol - [0904.3087v4] Distributed Maintenance of Anytime Available Spanning Trees in Dynamic Networks

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_fb3bd595.py", line 134, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_fb3bd595.py", line 96, in run_trial
    query = memoized_recursive_query_enumeration(lambda x: f[x], n)
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_fb3bd595.py", line 21, in memoized_recu
    @lru_cache(None)
     ^^^^^^^^^
NameError: name 'lru_cache' is not defined
```

Judge reasoning

The test crashed with a NameError ('lru_cache' is not defined) before producing any data, so no support fraction or slack values were computed. Per the rules, a crashed test yields INCONCLUSIVE. | next: Fix the test by importing lru_cache from functools (add 'from functools import lru_cache') and rerun the full enumeration over n∈{1,2,3}, b∈{2,3}.

Doob Martingale Variance Gap Predicts Worst-Case DPLL Depth

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Doob martingale concentration / Efron-Stein variance decomposition on random CSP exposures × DPLL search-tree depth on random 3-CNF (average-case to worst-case lift)
- **Recorded**: 2026-04-27 07:44 UTC
- **Entry ID**: `6487f7d68c31`

Statement

Let F ~ F(n, m=4n) be a random 3-CNF and let X(F) = log2(DPLL_leaves(F)) under fixed unit-propagation + first-unassigned variable order. Expose clauses one at a time to form the Doob martingale M_i = E[X | C_1,…,C_i] and let V(F) = sum_i (M_i - M_{i-1})^2 be its realized quadratic variation. Then for every n in [10,20] there exist constants c1,c2 > 0 (independent of n) with c1 * sqrt(V(F)) * log n <= max_{F' agreeing with F on all but one clause} X(F') - E[X] <= c2 * sqrt(V(F)) * log n with probability >= 1 - 1/n; equivalently, the average-case Doob variance V(F) tightly predicts the one-clause-perturbation worst-case depth gap.

Rationale

Efron-Stein variance bounds give average-case concentration, but the lower-tail (one-clause-resampling worst-case) is rarely tied to the same martingale variance — yet Impagliazzo-Wigderson-style hardness amplification suggests average-case fluctuation should control nearby worst-case spikes. If the two-sided bound holds, an average-case lower bound on V translates into a worst-case DPLL depth bound via a single resampling step, a concrete avg-to-worst lift on a small computable scale.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [1411.0186v2] Algorithmic randomness for Doob's martingale convergence theorem in continuous time - [1408.3470v1] Efron-Stein Inequalities for Random Matrices - [1908.05361v2] Placing quantified variants of 3-SAT and Not-All-Equal 3-SAT in the polynomial hierarchy - [0506069v1] A generating function method for the average-case analysis of DPLL - [2002.02797v4] Variational Depth Search in ResNets

Empirical Test

- exit code: `0`, elapsed: `0.04s`


```
{"TRIAL": {"seed": 11, "metric_name": "Doob Variance Gap Predicts Worst-Case DPLL Depth", "metric_valu
{"TRIAL": {"seed": 23, "metric_name": "Doob Variance Gap Predicts Worst-Case DPLL Depth", "metric_valu
{"TRIAL": {"seed": 37, "metric_name": "Doob Variance Gap Predicts Worst-Case DPLL Depth", "metric_valu
{"TRIAL": {"seed": 53, "metric_name": "Doob Variance Gap Predicts Worst-Case DPLL Depth", "metric_valu
{"TRIAL": {"seed": 71, "metric_name": "Doob Variance Gap Predicts Worst-Case DPLL Depth", "metric_valu
RESULT: SUPPORTED mean=0.0 std=0.0 support_fraction=1.0
```


Judge reasoning

The metric_value is exactly 0 with zero variance across all 5 seeds and only 6 instances per seed, indicating the test likely checked a trivially-satisfiable existential bound rather than the pre-registered OLS R^2/slope/window-fraction criterion. No evidence of pooled R^2, fitted slope, or per-n window fractions was reported, so the support condition is not unambiguously met. | next: Re-run with seeds 1..200 across n in {10,12,14,16,18,20} and explicitly compute pooled OLS regression of log(gap